Optimization & Uncertainty in Learning Report Rubric
CS7641: Machine Learning
Fall 2026

Schedule

- **10/19/2026 (Monday):** OL Grading Begins & Overview of Rubric
- **10/21/2026 (Wednesday):** OL First 2 Calibration
- **10/23/2026 (Friday):** OL Calibration with Examples and Head TAs Check
- **10/27/2026 (Tuesday):** OL 33–50% Completed
- **10/30/2026 (Friday):** OL 50–80% Completed
- **11/3/2026 (Tuesday):** OL 100% Completed
- **11/4/2026 (Wednesday):** OL Posted

Total: 100 points + 5 extra credit

This rubric is aligned to the Fall 2026 Optimization & Uncertainty in Learning (OL) assignment specification. The scoring categories and points remain unchanged from the prior OL rubric.

How to Use This Rubric

This rubric is designed to support **consistent grading across TAs** while still rewarding strong scientific work.

Each line item has a **fixed point value**. This is an all-or-nothing rubric. Either the requirement is clearly met and the points are awarded, or it is not met and the points are not awarded. Do not award partial credit within a listed item unless that item explicitly separates points into smaller sub-items.

Award points **only** when the evidence is explicit, correct, and directly supported by the report. If the report suggests a requirement was met but does not clearly demonstrate it, treat the item as not satisfied.

The reproducibility sheet (REPRO) may contain extended heatmaps, full seed tables, detailed regularization sweeps, additional diagnostics, environment details, and implementation details. However, the main report must still stand alone as a formal technical report. Students should not move all substantive interpretation into REPRO.

Avoiding Double Jeopardy

Do not penalize the same issue multiple times unless it affects multiple independent requirements.

For example, if a student omits GA in Part 1, remove the points tied to missing RO algorithm completion, operator disclosure, curves, or interpretation where appropriate. Do not also apply a separate global penalty unless the omission makes the entire assignment impossible to evaluate.

TA pre-grading checks identify non-scoring conditions, grading boundaries, access checks, or escalation cases. They should not create additional point deductions beyond the rubric. A temporary zero for unverifiable required artifacts is an administrative hold, not a separate penalty; restore the earned rubric score once the required access is verified.

Minimum Standard for Experimental Exploration

When a rubric item requires "multiple values," the default expectation is at least **three distinct and reasonable values** for the relevant scalar hyperparameter unless the student gives a clear computational justification for a smaller but still meaningful sweep.

A reasonable sweep should cover a range broad enough to reveal underfitting, overfitting, instability, or performance saturation. Testing trivially different learning rates or regularization values does not count as meaningful exploration unless justified by prior evidence.

For computationally expensive experiments, a smaller sweep may be acceptable if the student explains the computational constraint, reports runtime evidence, and still compares enough settings to support the final claim.

Required Task

Students must complete the required task exactly as follows:

- **Dataset:** Diabetes Health Indicators Dataset derived from the 2015 Behavioral Risk Factor Surveillance System (BRFSS), studying General Health.
- **Required file:** `diabetes_012_health_indicators_BRFSS2015.csv`
- **Do not accept:** `diabetes_binary_health_indicators_BRFSS2015.csv` or `diabetes_binary_5050split_health_indicators_BRFSS2015.csv`.
- **Task:** Five-class multiclass classification.
- **Target:** `GenHlth`, with 1 = Excellent, 2 = Very Good, 3 = Good, 4 = Fair, and 5 = Poor.
- **Full dataset size:** 253,680 instances.
- **Working sample:** Approximately 20,000 instances, using the same sample as the SL Report unless a change is explicitly justified.
- **Data and splits:** Reuse SL preprocessing and train/validation/test splits; disclose and justify material changes and qualify comparisons to SL if anything differs.
- **Backbone:** The PyTorch neural-network architecture finalized in SL, with the plain-SGD SL PyTorch model as the baseline. Do **not** use the scikit-learn MLP as the OL backbone.
- **Required metrics:** Validation loss, Accuracy, Macro-F1, balanced accuracy, confusion matrix, and per-class precision/recall/F1 discussion.

The `GenHlth` target must not be included among the predictor variables. The numeric ordering of the five response categories does not change the assignment into regression; evaluate predictions as five-class classification.

Algorithm Review for TAs

This assignment evaluates **training interventions** on the **same PyTorch backbone** students finalized in the SL Report. The goal is not simply to maximize Accuracy. The goal is to diagnose **speed-to-target**, **stability across seeds**, **sensitivity to hyperparameters**, **generalization**, **class-level behavior**, and **compute trade-offs** under fixed budgets.

The BRFSS dataset includes binary health indicators, ordered categorical responses, and variables on different numeric scales. The five self-reported General Health categories are not equally represented, and adjacent health categories may be difficult to distinguish. Scaling, feature encoding, optimization, and regularization can therefore affect both convergence and class-level performance. Overall Accuracy may hide weak recall for less-represented Fair or Poor classes, so Macro-F1, balanced accuracy, confusion matrices, and per-class precision/recall/F1 matter.

The expected result is not one universal optimizer ranking. A strong report may find that Adam improves speed but not test Macro-F1, that regularization improves seed stability while hurting a particular health class, or that RO fine-tuning improves validation loss only marginally relative to its compute cost. TAs should reward controlled comparisons and defensible interpretation, not just higher final metrics.

General Plausible Behavior on BRFSS General Health

TAs should expect trade-offs rather than a clean optimizer hierarchy. The approximately 20,000-instance working sample may be sufficient to reveal stochastic training differences, yet less-represented classes can yield variable per-class estimates. A method can improve validation loss with little change in Macro-F1; increase overall Accuracy while degrading Fair or Poor recall; or improve balanced accuracy while reducing overall Accuracy. These are all plausible outcomes and must be interpreted from the student's evidence.

Do not grade toward a required performance threshold. Results in approximately the 40–60% range can be entirely reasonable for this multiclass BRFSS task, depending on the metric, baseline, data split, model, and training setup. Neither modest scores nor exceptionally strong scores should be judged without checking baselines, leakage, curves, class-level behavior, and compute.

The most important grading question is not "Did this method win?" but "Did the student explain what changed, under what budget, and why the change is credible?" A high-performing result with unclear compute accounting, missing seed dispersion, or inconsistent splits is weaker than a modest result with clean controls and honest interpretation.

TAs should distinguish:

- **optimization improvement:** faster reduction in validation loss or smoother convergence;
- **generalization improvement:** improved held-out test Macro-F1, balanced accuracy, or class-level behavior;
- **stability improvement:** lower variability across seeds or fewer divergent runs;
- **efficiency improvement:** similar performance with fewer gradient updates, function evaluations, or wall-clock time;
- **class-level improvement:** better precision, recall, or F1 for difficult health categories, even if Accuracy changes little.

Strong reports separate these claims. Weak reports collapse them into "Adam performed best" or "regularization helped."

Part 1: Randomized Optimization

Students apply Randomized Hill Climbing (RHC), Simulated Annealing (SA), and Genetic Algorithm (GA) to the final 1–3 layers of the PyTorch BRFSS General Health network. They must freeze earlier layers, use an approximately $\leq 50{,}000$-parameter trainable RO budget, and report the exact count.

RO must optimize a deterministic validation-loss objective. Evaluation should use `model.eval()` so dropout is off and normalization layers, if any, use evaluation behavior. Every candidate objective call counts as a function evaluation. If a fixed validation subset is used for computational reasons, its size, seed, and sampling method must be disclosed, it should preserve class representation as appropriate, and it must be separate from the test set.

Plausible RO behavior

RO may produce small gains, no gains, or unstable gains. This is not automatically failure. Search-based methods over neural-network weights can be inefficient in high-dimensional spaces even when restricted to final layers. Strong work can report that RO plateaued or underperformed and explain it through parameter count, perturbation scale, function-evaluation budget, validation-loss trajectories, and wall-clock time.

Plausible outcomes include:

- **RHC improves quickly then plateaus.** Small perturbations may find nearby improvements but not escape a basin. Analysis should connect this to step size, restart policy, and best-so-far curves.
- **SA accepts worse candidates before improving.** Early exploration is plausible under an appropriate temperature schedule. Aggressive cooling may resemble RHC; slow cooling may consume compute without corresponding improvements.
- **GA is noisy and expensive.** Population-based evaluation may consume many objective calls, making apparently strong per-generation progress expensive per evaluation or wall-clock time.

- **RO improves validation loss but not Macro-F1.** Final-layer changes may adjust confidence without changing many predicted classes, especially if predictions remain concentrated in common General Health categories.
- **RO improves the validation objective but harms test performance.** A small or repeatedly reused validation subset can induce selection bias, especially for less-represented health categories.
- **RO is dominated by gradient training.** This is legitimate if the evidence shows SGD or Adam reaches a comparable or better objective using less compute.

TAs must not require RO to beat Adam or the SL baseline. Require an evidence-based explanation of whether RO provided useful fine-tuning, robustness, or insight relative to its computational cost.

RO behaviors that should raise concern

- RO greatly outperforms all gradient methods without explanation, seed checks, or leakage controls.
- RO operates on the full network but is described as final-layer tuning.
- The objective is evaluated while dropout is active or another stochastic training behavior changes candidate fitness.
- Function evaluations are counted inconsistently across RHC, SA, and GA.
- GA is compared to RHC or SA by generations rather than objective calls.
- The report shows only the best seed and hides failed or divergent RO runs.

Part 2: Adam-Family Ablations

Students train the full PyTorch backbone using SGD without momentum, SGD with momentum, Nesterov momentum, Adam, Adam without bias correction, Adam with $\beta_1 = 0$, and AdamW.

The study concerns optimizer ingredients. Students should explain whether momentum, Nesterov acceleration, adaptive scaling, bias correction, or decoupled weight decay affected convergence, validation loss, seed stability, or class-level performance, rather than merely naming the highest-scoring optimizer.

Plausible Adam-family behavior

Adam-family methods may reduce validation loss faster than SGD, especially early in training, but faster optimization does not guarantee better test Macro-F1 or balanced accuracy.

Plausible outcomes include:

- **SGD starts slowly but is competitive by the budget limit.** Appropriate learning rates and sufficient updates may reduce an early gap.
- **Momentum improves SGD.** Momentum can accelerate descent in persistent directions and smooth mini-batch noise.
- **Nesterov produces small or inconsistent gains.** Nesterov and classical momentum may be close depending on the chosen settings and budget.
- **Adam reaches the validation-loss threshold fastest.** Adaptive scaling may help when gradients or feature scales differ, but the speed advantage need not improve test class-level metrics.
- **Adam without bias correction behaves differently early.** Biased moment estimates change the early effective update scale; the effect depends on the implementation and learning rate.
- **Adam with $\beta_1 = 0$ behaves like an RMSProp-like condition.** Second-moment scaling remains while first-moment smoothing is removed.
- **AdamW improves generalization.** Decoupled weight decay may help control weights, but must be distinguished from coupled L2 regularization in standard Adam.

- **Adam improves validation loss but not class-level predictions.** Changes in confidence need not improve discrimination between neighboring General Health categories.
- **The ranking depends on the metric.** The best loss, Macro-F1, balanced accuracy, and speed-to-threshold may belong to different methods.

TAs should reward reports distinguishing convergence speed, final validation loss, test generalization, and class-level behavior. Ranking optimizers by final Accuracy alone is inadequate.

Sensitivity heatmap behavior

Sensitivity heatmaps over (α, β_1) and (α, β_2) should reveal stable, unstable, and sensitive regions rather than merely identify a single best cell. Coarse grids are acceptable, and extended versions may appear in REPRO.

Plausible patterns include:

- **A broad stable region:** robust nearby settings.
- **A narrow good region:** stronger tuning sensitivity.
- **Divergence at high learning rates:** expected in some conditions and must be marked.
- **Weak sensitivity to one beta:** potentially dominated by learning rate or limited training duration.
- **Interaction effects:** one beta value changes which learning rates remain stable.

Adam-family behaviors that should raise concern

- AdamW is reported as the standard Adam baseline without distinguishing its decay rule.
- Optimizers use unexplained differences in architecture, batch size, preprocessing, or splits.
- Different initializations are used without a clear, fair seed protocol.
- Divergent runs disappear from the summaries.
- The threshold ℓ^* is selected using test results.
- Heatmap cells receive inconsistent compute budgets without justification.
- Conclusions rely only on a single favorable seed.

Part 3: Regularization Study

Students use **standard Adam** with the best Part 2 hyperparameters, not AdamW, and must not retune the Adam optimizer inside Part 3. Only regularization mechanisms should change.

Students study L2 weight decay, early stopping, dropout, and noise-based regularization, including label smoothing and modest input noise or augmentation. They compare standard Adam without added regularization, the best single regularizer, and the best combination under matched budgets.

Plausible regularization behavior

Regularization can improve or worsen generalization depending on magnitude, interactions, and baseline model capacity. Analyze effects through validation loss, train-validation gaps, test Macro-F1, balanced accuracy, class-level behavior, and seed stability.

Plausible outcomes include:

- **L2 weight decay helps until it becomes too strong.** Excessive decay can underfit and lower both training and validation performance.
- **Early stopping limits late-epoch overfitting.** Reports should identify monitored metric, patience, maximum budget, and restore-best behavior.

- **Dropout helps selectively or hurts a compact tabular network.** Added stochasticity is not automatically beneficial.
- **Label smoothing improves loss or confidence behavior without improving Macro-F1.** Hard predictions may remain unchanged.
- **Input noise helps robustness or harms class separation.** BRFSS contains binary and ordered categorical variables. Modest noise on genuinely continuous variables can be defensible; indiscriminate additive noise on discrete indicators or ordered categories can create invalid feature values. Students should state which features were perturbed, how, and why the transformations preserve meaningful inputs. A justified discrete-feature augmentation is not automatically disallowed.
- **The best combination is not necessarily every regularizer together.** Over-regularization can erase gains from individual methods.
- **Minority-class recall improves while Accuracy declines.** This may be an appropriate trade-off when supported by Macro-F1 or balanced accuracy.
- **Seed variability declines without a large mean-score change.** Stability may itself be a meaningful finding.

TAs should reward explanations of *how* regularization changed behavior. Strong reports compare these effects with optimizer changes, for example optimizer speed-to-threshold versus regularization effects on generalization gaps or difficult General Health categories.

Regularization behaviors that should raise concern

- Part 3 silently changes Adam learning rate, betas, batch size, or optimizer family.
- AdamW is used instead of standard Adam.
- Dropout placement, rate, or early-stopping policy is not documented.
- Early stopping or hyperparameter selection uses the test set.
- Label smoothing or input noise intended for training is also applied during validation or testing.
- Noise creates invalid discrete feature values without explanation or a defensible augmentation scheme.
- The best combination is selected on the test set.
- Generalization improvements are claimed using only training loss.

Main Report versus REPRO

The main report should contain compact summaries, key figures, and interpretation. Extended heatmaps, full seed tables, detailed regularization sweeps, full logs, and additional diagnostic plots may appear in REPRO. TAs should not require every diagnostic in the 8-page report, but must require compact evidence supporting its central claims.

For example, claims that "Adam was more stable," "dropout improved recall for Poor health," or "RO was not worth the compute" must be supported by a summary in the main report. REPRO can supply detailed underlying tables, seeds, and logs.

Rubric

Total Points: 100 + 5 Extra Credit

TA Start Check

Complete these checks before scoring each submission. These checks identify access issues, grading boundaries, or escalation cases; they do not independently add or remove scored rubric points.

☐ Both required PDFs, `OL Report` and `REPRO OL`, are present and readable.

☐ The student uses `diabetes_012_health_indicators_BRFSS2015.csv`, not a binary diabetes file.

☐ `GenHlth` is the five-class target (classes 1–5) and is not included among predictors.

☐ The approximately 20,000-instance SL working sample, split, and preprocessing are reused or deviations are documented.

☐ The OL backbone is the SL PyTorch model, with the SGD-only SL baseline identified.

☐ A GT Enterprise Overleaf **Edit Link** is provided and accessible.

☐ The GT Enterprise GitHub repository and final commit hash are accessible.

☐ Reproducibility instructions, seed information, and compute accounting are present.

☐ The AI Use Statement is present.

☐ Check if the student is using Blue/Yellow Graphs or the word flummoxed.

☐ The main report is within the 8-page scoring limit and follows the IEEE conference format.

☐ The report reads as a formal technical paper rather than a checklist or bullet-only draft.

Check 1: Readability

Confirm that the submitted report PDF can be opened and evaluated. If the report cannot be opened or evaluated at all, assign a zero and document the issue. For a missing or inaccessible required supplemental artifact that might be corrected, follow the submission-verification procedure below.

Check 2: Required BRFSS General Health Task

Confirm that the required 2015 BRFSS source file is used and that `GenHlth` is treated as a five-class classification target. Confirm that the student has not instead used `Diabetes_012` as the target, used a binary diabetes file, collapsed `GenHlth` into two classes, or performed regression.

Action: Use of the wrong required dataset/file or target is a substantive failure of the assignment and receives a zero, rather than a temporary access hold.

Check 3: Submission Access

Confirm the following:

- An **Overleaf Edit Link** is available in REPRO, the main report, or the Canvas submission comment and points to the student's Georgia Tech / Enterprise Overleaf project.
- The Overleaf project can be accessed by staff with the required edit-level access. A read-only View Link alone does not meet the Fall 2026 requirement.
- The submitted final commit hash identifies code in a **GT Enterprise GitHub repository**, not a personal GitHub repository, and staff can access it.

- The assignment includes the two required PDFs and sufficiently usable reproduction instructions.

Action: If a required artifact or access condition is missing or unverifiable, continue grading available work, provide substantive feedback, record the earned rubric score, and enter a **temporary overall zero**. Identify the missing item and escalate access questions to the Head TA as appropriate. Once the item is supplied and verified, restore the recorded earned score. Do not add another deduction for the same access issue.

Check 4: Page Limit

Do not read or score content beyond page 8 of the main report, including any references beyond this limit. Content appearing only after the limit is considered missing when applying the scored rubric. The separate REPRO PDF can contain allowed supplementary technical details but does not replace main-report analysis.

Check 5: Draft Format

Confirm that the report follows the IEEE conference template and reads like a formal technical paper. Bullets are permissible for compact protocols, hyperparameters, and reproducibility details, but paragraph-based reasoning is expected in Results, Discussion, and Conclusion. Multiple instances of four or more consecutive bullets in analysis-heavy sections can indicate excessive checklist-style writing unless justified by the content.

Action: Treat excessive bullet-only writing that substitutes for scientific interpretation as a draft-quality report. Apply the scored analytical criteria to the evidence actually provided; do not invent an unlisted blanket point penalty. Escalate ambiguous draft-format cases to Head TAs for consistent treatment.

Integrity and Provenance Triage

Overleaf history, repository history, AI-use disclosures, Turnitin overlaps, AI-detector indicators, and agent-generated files can inform a targeted review, but no single one establishes misconduct. Especially useful checks concern **inconsistencies** between reported methods, actual code, figures, numerical results, citations, and stated AI use.

Examples worth escalating include unsupported claims about runs that do not exist, numbers that contradict tables, fabricated references, report-ready AI output presented as original analysis, or code that does not implement the named optimizer variant. Generic style, polished prose, common phrasing, a single large paste, and high AI-detector scores are not sufficient evidence by themselves.

Action: Grade academic work normally where possible, document the concrete concern, preserve the relevant evidence, and refer it to the Head TA/instructor. Do not impose ad hoc misconduct deductions or accuse students in routine grading feedback.

Rubric Points

Grade all items below **all-or-nothing**. If an item is worth 2 points, award either 0 or 2 points. Use the "Award if" and "Do not award if" descriptions as calibration guidance, not as a demand for identical wording.

Compliance (16 points)

Data Compliance (3 points)

- **(2 pts) BRFSS sample.**
  Award if: The report uses an approximately 20,000-instance BRFSS General Health working dataset matching the SL Report sample, or explicitly documents and justifies any change while explaining its effect on SL comparability.
  Do not award if: The student uses the full 253,680-instance dataset without justification, an arbitrarily small or substantially different sample, or does not explain a sample change.

- **(1 pt) Target and SL backbone.**
  Award if: The report uses the required BRFSS file, identifies `GenHlth` classes 1–5 as the multiclass target,

excludes it from predictors, and uses the PyTorch SL neural-network backbone as the OL model.
Do not award if: The source file or target is wrong, the label is collapsed or treated as regression, the target leaks into predictors, or the PyTorch SL backbone is not identifiable.

Submission Access (4 points)

- **(2 pts) Enterprise Overleaf.**
Award if: A working Overleaf **Edit Link** to the student's Georgia Tech / Enterprise project is included in the required submission materials and is accessible to course staff.
Do not award if: The link is missing, broken, read-only, inaccessible, or tied only to a personal Overleaf account. Apply the temporary-zero verification process rather than an additional access penalty.

- **(2 pts) Enterprise GitHub.**
Award if: The report or REPRO materials include a valid final commit hash from the student's GT Enterprise GitHub repository, and course staff can access the repository and commit.
Do not award if: The hash is missing, invalid, inaccessible, or points only to a personal GitHub repository. Apply the temporary-zero verification process rather than an additional access penalty.

Reproducibility (3 points)

- **(2 pts) Run instructions.**
Award if: The student provides clear commands, environment and package requirements, data paths, execution order, and random seeds sufficient to reproduce outputs on a standard Linux machine.
Do not award if: Instructions are missing or insufficient to reproduce the experiments, such as saying only "run the notebook" without environment, data, or execution details.

- **(1 pt) Experiment details.**
Award if: REPRO includes the SL-to-OL sample, preprocessing and split correspondence; optimizer and RO operator settings; regularization values; seeds; compute accounting; metric calculations; and enough information to recreate the main figures and tables.
Do not award if: Key experiment details are missing, vague, not linked to the submitted results, or insufficient to verify the claimed SL baseline comparison.

Figure Legibility (2 points)

- **(2 pts) Readable figures and tables.**
Award if: Figures, tables, axes, legends, captions, heatmap labels, and text are readable at normal grading zoom, and visual evidence can be assessed without guesswork.
Do not award if: Multiple figures or tables are too small, clipped, blurry, poorly labeled, or otherwise difficult to evaluate reliably.

Attribution (4 points)

- **(2 pts) Outside sources.**
Award if: At least two relevant peer-reviewed sources beyond course materials are used to justify methods or interpret results, with IEEE-style citations in the report body.
Do not award if: Fewer than two qualifying sources are used, references are irrelevant or listed without use, or only textbooks, blogs, documentation, tutorials, lecture slides, or non-peer-reviewed preprints are supplied.

- **(2 pts) AI usage statement.**
Award if: The report includes a clear AI Use Statement identifying the tools used, how they assisted, and affirming that the student reviewed, verified, and understood all assisted content. A statement of no AI use is sufficient if that is accurate.
Do not award if: The statement is missing, vague, does not identify tools or assistance, or fails to affirm review and understanding of assisted material.

Study Design (10 points)

Goal: The report should establish a controlled experimental design before interpreting optimization and regularization results, then revisit its hypothesis using evidence.

- **(2 pts) Splits and leakage controls.**
  Award if: The report states the reused SL train/validation/test splits, uses the held-out test set only for final evaluation, and confirms learned preprocessing is fit only on training data or training folds. Any material SL-to-OL change is justified.
  Do not award if: Splits are ambiguous or differ without explanation, test data are used for tuning, or preprocessing or target leakage is present or unresolved.

- **(2 pts) Compute and seed protocol.**
  Award if: The report defines gradient evaluations, function evaluations, wall-clock measurement, random-seed protocol, hardware class, and batching assumptions consistently enough to support fair comparisons.
  Do not award if: Compute accounting is missing or inconsistent, or methods are compared only by epoch count without explaining updates, evaluations, or wall-clock.

- **(3 pts) Early hypothesis.**
  Award if: At least one hypothesis is stated before full results and predicts optimization or regularization behavior using course concepts such as momentum, adaptive scaling, bias correction, regularization, stability, or compute trade-offs.
  Do not award if: The hypothesis is generic, appears after the results, predicts only a winner without theoretical reasoning, or is unrelated to OL behavior.

- **(3 pts) Hypothesis resolution.**
  Award if: The original hypothesis is explicitly accepted, rejected, or qualified using quantitative evidence from at least two of the following: validation loss, test Macro-F1, balanced accuracy, class-level behavior, seed stability, gradient evaluations, function evaluations, or wall-clock time.
  Do not award if: The hypothesis is not revisited, is discussed only vaguely, or is resolved without quantitative evidence from the reported experiments.

Randomized Optimization (18 points)

Goal: Evaluate whether search-based methods can improve or stabilize final-layer behavior under a strict parameter and compute budget.

- **(3 pts) Layer scope and parameter cap.**
  Award if: The report identifies exactly which final 1–3 layers are unfrozen, confirms earlier layers are frozen during RO, reports the exact trainable RO parameter count, and keeps the count approximately $\leq 50{,}000$ or provides clear computational justification for an allowed deviation.
  Do not award if: RO runs on the full network, the unfrozen layers or frozen status are unclear, or the trainable parameter count is missing.

- **(3 pts) Required RO algorithms completed.**
  Award if: RHC, SA, and GA are all run on the same final-layer optimization problem using the same validation objective, data split, model backbone, and comparable initialization or starting-point protocol.
  Do not award if: Any required algorithm is missing, the layer scope or objectives differ, or starting conditions make the comparison unfair.

- **(3 pts) Operator settings and perturbation design.**
  Award if: The report reproduces each RO method's settings: RHC restarts and step-size policy; SA initial temperature, cooling schedule, and any step-size cooling; GA population size, selection, crossover, mutation, elitism, and generation count; and perturbation distribution, scale, and adaptation schedule where relevant.
  Do not award if: Algorithms are named without operator details, perturbation scale/distribution is omitted, or the process generating candidate weights is unclear.

- **(3 pts) Deterministic validation objective and function-evaluation accounting.**
  Award if: RO optimizes validation loss using deterministic evaluation behavior such as `model.eval()`, every candidate objective call is counted as a function evaluation, and any fixed validation subset is disclosed with size, seed, and sampling method.

Do not award if: The objective changes mid-run, stochastic dropout or other training behavior affects candidate evaluation, function evaluations are inconsistent, or the test set is used as the RO objective.

- **(3 pts) RO progress curves and budget matching.**
  Award if: The report includes best-so-far validation objective versus function evaluations for RHC, SA, and GA; reports wall-clock time; and compares methods under matched or clearly justified evaluation budgets. GA is compared by objective calls rather than generation counts alone.
  Do not award if: Progress curves are missing, axes or budgets are incomparable without explanation, GA is compared by generation count alone, or only final objective values are shown.

- **(3 pts) RO interpretation and final evaluation.**
  Award if: The report explains whether RO provided meaningful tail improvement, plateaued, stalled, or overfit validation behavior using validation loss, evaluations, time, final test Accuracy, Macro-F1, balanced accuracy, and class-level behavior. It compares RO with the SL PyTorch SGD baseline or gradient-based methods.
  Do not award if: The report only names the best RO method, ignores compute cost, omits final multiclass evidence, or never discusses whether the RO result justified the compute.

Adam-Family Ablations (20 points)

Goal: Dissect which optimizer ingredients affect training speed, stability, generalization, and class-level behavior.

- **(4 pts) Seven optimizer variants.**
  Award if: The report runs all required optimizer variants on the full PyTorch backbone: SGD without momentum, SGD with momentum, Nesterov momentum, Adam, Adam without bias correction, Adam with $\beta_1 = 0$, and AdamW. For each optimizer it states learning rate, momentum or beta values, epsilon where relevant, weight decay behavior, batch size, maximum budget, and any schedule.
  Do not award if: Any required optimizer is missing, replaced, mislabeled, not actually run, or not documented well enough to verify the ablation.

- **(3 pts) Fair optimizer protocol.**
  Award if: The comparison uses the same backbone, split, preprocessing, batch size unless justified, budget definition, metric reporting, and initialization strategy. Paired seeds map to comparable initial weights across methods. Learning-rate schedules, clipping, early stopping, and other training-loop changes are disclosed.
  Do not award if: Methods use incompatible budgets, initialization, schedules, splits, preprocessing, or architecture without justification.

- **(3 pts) Loss-vs-compute and threshold analysis.**
  Award if: The report shows validation loss versus wall-clock time and/or gradient evaluations, defines a fixed validation-loss threshold ℓ^* using training/validation evidence only, and reports time or steps to reach ℓ^*. Optimizers that do not reach the threshold remain in the analysis and are marked as exceeding the budget.
  Do not award if: The threshold is missing, changes by optimizer, is selected using test performance, excludes failures, or lacks loss-versus-compute evidence.

- **(3 pts) Sensitivity heatmaps or summaries.**
  Award if: The report provides compact sensitivity heatmaps or summaries for coarse (α, β_1) and (α, β_2) grids, uses identical budgets per grid point, marks divergent or failed cells, and interprets stable, unstable, and sensitive regions. Extended heatmaps may appear in REPRO.
  Do not award if: Sensitivity analysis is missing, budgets differ without justification, failed cells are hidden, or heatmaps are presented without meaningful interpretation.

- **(3 pts) Seed stability and failure reporting.**
  Award if: The main optimizer comparison uses at least 3 seeds unless computationally justified; reports dispersion such as median/IQR, variance bands, or standard deviation; and includes divergent or failed runs in its stability discussion.
  Do not award if: Only one seed supports the main claims, failed runs are hidden, dispersion is absent, or conclusions rely on one best-seed result.

- **(4 pts) Generalization and optimizer-ingredient interpretation.**
  Award if: The report compares training and validation behavior; final test Accuracy, Macro-F1, balanced accuracy, and class-level results; and interprets what momentum, Nesterov acceleration, adaptive scaling,

bias correction, and decoupled weight decay actually changed on BRFSS General Health.
Do not award if: Only final test Accuracy is reported, train-validation behavior or class-level effects are ignored, or optimizer variants are ranked without explaining the ingredients responsible for observed differences.

Regularization Study (16 points)

Goal: Measure how regularization affects BRFSS General Health generalization under standard Adam and a fixed compute budget.

- **(3 pts) Standard Adam fixed from Part 2.**
  Award if: The report uses standard Adam with the best Part 2 hyperparameters and does not retune Adam settings inside Part 3. Learning rate, betas, epsilon, batch size, and budget remain fixed unless a necessary deviation is explicitly justified. Only regularization mechanisms change.
  Do not award if: The student switches to AdamW, changes optimizer family, retunes the learning rate or betas in Part 3, changes batch size without justification, or changes the backbone without justification.

- **(4 pts) Required regularizers tested and disclosed.**
  Award if: The report tests and discloses L2 weight decay, early stopping, dropout, label smoothing, and modest input noise or augmentation. It states weight-decay values, early-stopping monitored quantity and patience, whether best weights are restored, dropout layer placement and rates, label-smoothing magnitude, input-noise or augmentation policy and magnitude, and confirms all training-only interventions are confined to the appropriate split and feature types.
  Do not award if: A required technique is missing, inadequately documented, applied to validation/test data inappropriately, or introduces invalid BRFSS indicator/category values without explanation of a valid augmentation approach.

- **(3 pts) Budget-matched regularization comparison.**
  Award if: The report compares standard Adam with no added regularization, the best single regularizer, and the best regularization combination under matched compute budgets and a consistent seed protocol. If early stopping reduces epochs, it separates saved compute from different training opportunity.
  Do not award if: Budgets differ without justification, the baseline/single/combination comparison is missing, test performance selects the best condition, or only favorable single runs are reported.

- **(2 pts) Selection evidence and REPRO support.**
  Award if: The main report includes compact evidence showing how regularizer settings were selected, such as a small sweep summary, validation-loss comparison, or metric table. Detailed sweeps, full seed tables, and additional plots are available in REPRO when too large for the main report.
  Do not award if: Only a final regularized result is provided without explaining its selection, or referenced selection evidence is missing from both the report and REPRO.

- **(4 pts) Interpretation versus optimizer changes.**
  Award if: The report analyzes whether regularization improved validation loss, Macro-F1, balanced accuracy, Fair/Poor or other class-level recall, train-validation gaps, seed stability, or test performance; and compares the magnitude and nature of the effects with Part 2 optimizer changes.
  Do not award if: Results are listed without explaining the changes, generalization improvement is claimed from training loss alone, or effects are not compared with optimizer changes.

Discussion (20 points)

Goal: The Discussion should move beyond presenting results and explain what the OL study reveals about training neural networks on BRFSS General Health. This section should synthesize RO, Adam-family ablations, and regularization rather than treating them as isolated checklists.

The Discussion should be written in paragraph form. It should not simply repeat a table, provide a bullet-only recap, or summarize each optimizer separately. It should explain what the results mean, why they likely occurred, how reliable they are, and what a practitioner should do with the evidence.

Expected discussion content

A strong Discussion should address these ideas, though students may organize them differently:

- which intervention primarily improved optimization speed;
- which intervention primarily changed generalization or class-level behavior;
- which intervention improved stability across seeds;
- whether improvements in validation loss transferred to test Macro-F1, balanced accuracy, or difficult-class recall;
- whether a performance gain justified its additional compute;
- whether the preferred technique depends on which metric matters most;
- what these results imply specifically for a PyTorch neural network on BRFSS General Health.

Limitations expectation

The Discussion must include a concrete limitations analysis. Useful examples include the approximately 20,000-instance sample, imbalanced General Health classes, class overlap, uncertainty for less-represented Fair/Poor groups, limited RO function-evaluation budgets, final-layer-only RO, coarse Adam heatmaps, limited seeds, validation-subset use during RO, hardware constraints, dependence on the SL backbone, or uncertainty arising from input encodings and discrete-feature regularization. These should be connected to specific evidence; "more tuning could be done" alone is insufficient.

- **(4 pts) Cross-part synthesis.**
  Award if: The discussion compares RO, Adam-family ablations, and regularization, and explains how their effects differed using validation loss, test metrics, class-level behavior, stability, and compute. It distinguishes questions of tail tuning, optimizer dynamics, and generalization control.
  Do not award if: The report treats each part independently without cross-part synthesis, or declares one part best without identifying the nature of its improvement.

- **(4 pts) Compute-performance trade-offs.**
  Award if: The discussion explains whether gains justified their cost, using wall-clock time, gradient evaluations, function evaluations, and final metrics. It distinguishes inexpensive improvements from costly marginal gains and explains the compute value of RO, optimizer variations, and regularization.
  Do not award if: The report declares a winner without considering compute, lists compute without interpretation, or ignores large cost increases for small metric gains.

- **(3 pts) Optimization versus generalization versus stability.**
  Award if: The discussion separates convergence/validation-loss behavior from train-validation gaps, held-out test behavior, and seed variability. It identifies which aspect of training each method affected rather than treating these outcomes as interchangeable.
  Do not award if: The report equates lower loss, faster training, higher test metrics, and stability, or never explains which property improved.

- **(3 pts) Class-level and metric interpretation.**
  Award if: The discussion analyzes five-class General Health performance, including per-class precision, recall, and F1 or confusion patterns, and explains why Accuracy, Macro-F1, balanced accuracy, and validation loss may tell different stories for common and less-represented categories.
  Do not award if: Only aggregate scores are discussed, class-level behavior is ignored, or conflicting metric rankings are left unexplained.

- **(3 pts) Limitations and threats to validity.**
  Award if: The discussion identifies study-specific limitations and explains how they affect confidence in its conclusions, such as sample size, class imbalance, class overlap, limited RO budget, seed count, coarse sensitivity analysis, validation-subset use, architecture dependence, or preprocessing/feature constraints.
  Do not award if: Limitations are absent, generic, disconnected from results, or phrased only as a need for more tuning without evidence.

- **(3 pts) Practical decision rule.**
  Award if: The report ends with an evidence-based rule for BRFSS General Health describing when to prioritize optimizer changes, regularization, or RO final-layer fine-tuning. The rule is metric-aware, compute-aware, and acknowledges the limits of transfer to other datasets.
  Do not award if: The conclusion merely restates scores, recommends generic tuning or "use Adam" without distinguishing use cases, or proposes a rule unsupported by the experiments.

Extra Credit: Integrated Best Combination (+ up to 5 points)

Optional. Extra credit combines the best components from Parts 1–3. It should be a composition of prior findings, not a new broad search.

Required components

The integrated recipe uses:

- the best **standard Adam** hyperparameters from Part 2;
- the best **regularization combination** from Part 3;
- the best **RO fine-tuning method** from Part 1 on the same final 1–3 layers.

Constraints

- Do not introduce new optimizers.
- Do not run fresh wide hyperparameter sweeps.
- Do not change the RO layer scope or the backbone architecture.
- Keep data splits, seeds, hardware class, and batch size consistent with earlier parts.
- Limit RO fine-tuning to $\leq 10\%$ of the Part 1 RO function-evaluation budget.
- Limit interaction exploration to ≤ 5 small trials.

How Extra Credit is Awarded

These are **alternative award levels**, not additive amounts. The maximum is +5 points.

- **+1 point: Minimal integration, correct and fair.**
  Award if the report presents one integrated run, reports the compute budget, and compares it against a relevant Part 2 or Part 3 baseline under fair conditions. Do not award if model scope changes, compute details are absent, or the comparison is invalid.
- **+3 points: Full integration with evidence.**
  Award if the report integrates the best Part 2 Adam settings, best Part 3 regularization combination, and best Part 1 RO method, and gives a clear before/after comparison with at least 3 seeds unless computationally justified. Do not award if a required component is missing or only the best seed is shown.
- **+5 points: Full integration plus compute trade-off analysis.**
  Award if the +3 requirements are met and the report also compares test metrics, gradient steps, RO function evaluations, wall-clock time, and stability where applicable; and explains whether the incremental gain justifies its cost. Do not award full extra credit without a substantive compute-value argument.

TA Reminders

- Do not grade toward an expected optimizer winner or a required performance threshold.
- Keep the SL sample, split, preprocessing, and PyTorch SGD-only baseline central to OL comparisons.
- Accuracy alone is not enough for this imbalanced five-class General Health task.
- Separate faster optimization from better generalization, greater stability, and computational efficiency.
- Check that RO objective evaluations, optimizer updates, and wall-clock measures are defined consistently.
- Do not reward tuning or early stopping based on the held-out test set.
- For input noise, check feature meaning: binary and ordered categorical values should not be indiscriminately perturbed into invalid values.
- Use the main report for substantive claims and REPRO for supporting details and complete logs.
- Treat mismatches among figures, prose, code, and citations as more informative for provenance than writing style or detector scores.
- Grade available work and use a temporary zero for unverified required access, not an additional rubric deduction.
- Use the all-or-nothing rubric and escalate unclear or exceptional cases for consistent calibration.

Version Control

- v1.0 – 10/16/2026 – TJL updated OL rubric for Fall 2026 BRFSS General Health assignment, grading cadence, and Enterprise submission verification. Scored point allocation retained.

Rubric written by Theodore LaGrow.